

Employees and Sectors

A beginner-friendly SQL project that demonstrates how to manage employees and departments using a relational database. The project focuses on table relationships, data retrieval, filtering, aggregation, and JOIN operations.

Project Overview

This project contains two related tables:

- **employees** – Stores employee information such as name, salary, address, and department.
- **sectors** – Stores department details.

The project demonstrates how to retrieve meaningful business insights using SQL queries.

Technologies Used

- SQL
 - MySQL
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Database Schema

Employees Table

Column	Description
Employee_ID	Unique employee ID
Employee_Name	Employee name
Salary	Employee salary
Address	Employee address
Department_ID	Department reference

Sectors Table

Column	Description
Department_ID	Unique department ID
Department_Name	Department name

Database Relationship

- One **Department** can have multiple **Employees**.
 - Each **Employee** belongs to one **Department**.
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SQL Queries Included

1. Display all employee details

```
SELECT * FROM employees;
```

2. Find the salary of specific employees

```
SELECT Employee_Name, Salary
FROM employees
WHERE Employee_Name IN ('kiran rao', 'arjun patel');
```

3. Calculate the average salary for each department

```
SELECT
    s.Department_Name,
    AVG(e.Salary) AS Average_Salary
FROM employees e
JOIN sectors s
ON e.Department_ID = s.Department_ID
GROUP BY s.Department_Name;
```

4. Find the department with the highest average salary and list its employees

```
SELECT
    e.Employee_Name,
    e.Salary,
    s.Department_Name
FROM employees e
JOIN sectors s
ON e.Department_ID = s.Department_ID
WHERE e.Department_ID = (
```

```
SELECT Department_ID
FROM employees
GROUP BY Department_ID
ORDER BY AVG(Salary) DESC
LIMIT 1
);
```

5. Find the highest-paid employee in each department along with their location

```
SELECT
    e.Employee_Name,
    e.Salary,
    s.Department_Name,
    e.Address
FROM employees e
JOIN sectors s
ON e.Department_ID = s.Department_ID
WHERE (e.Department_ID, e.Salary) IN (
    SELECT Department_ID, MAX(Salary)
    FROM employees
    GROUP BY Department_ID
);
```

SQL Concepts Covered

- SELECT Statement
 - WHERE Clause
 - IN Operator
 - Aggregate Functions (`AVG` , `MAX`)
 - GROUP BY
 - INNER JOIN
 - Subqueries
 - Filtering Data
 - Relational Database Concepts
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Project Structure

```
employees-and-sectors/  
|— employees_and_sectors.sql  
|— README.md
```

Learning Outcomes

By completing this project, you will learn:

- Database design using relational tables
 - Primary and Foreign Keys
 - SQL filtering techniques
 - Aggregate functions
 - GROUP BY
 - INNER JOIN
 - Nested (Sub) Queries
 - Retrieving business insights using SQL
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